Quantification: 699_HYP_HP_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	20
interior lacuna count	17
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2623.41	1403	3027.55	px^2
major_axis_length_px	lacuna shape	87.099	67.1123	52.4465	px
minor_axis_length_px	lacuna shape	42.1822	34.896	20.5754	px
aspect_ratio	lacuna shape	2.1526	1.7046	0.9474	unitless
eccentricity	lacuna shape	0.7999	0.8098	0.1517	unitless
solidity	lacuna shape	0.7114	0.7052	0.0887	unitless
roots_count	headline	9.8235	11	5.1019	count
ring_length_r30_px	headline	383.765	360	211.643	px
ring_length_r60_px	reported	692.823	604	426.814	px
owned_length_px	ownership-dependent	875.959	727.813	802.564	px
edge_count	ownership-dependent	47.6471	35	45.8011	count
mean_edge_length_px	ownership-dependent	19.3373	19.2782	2.7809	px
perimeter_px	normalised	337.356	234.25	222.27	px
ring_area_r30_px2	normalised	9655.35	8057	4613.75	px^2
ring_area_r60_px2	normalised	21215.4	19375	10030.5	px^2
in_frame_fraction_r30	normalised	0.994	1	0.0205	unitless
in_frame_fraction_r60	normalised	0.9799	1	0.0568	unitless
ring_density_r30	normalised	0.0409017	0.0425232	0.0155465	px^-1
ring_density_r60	normalised	0.0326258	0.0322581	0.011726	px^-1
roots_per_100px_perimeter	normalised	3.2788	3.4555	1.6568	per 100 px
ring_attached_length_r30_px	network v2	334.059	317	221.726	px
ring_attached_length_r60_px	network v2	526.235	473	410.897	px
ring_length_w_r30_px	network v2	432.121	413.99	237.245	px
ring_length_w_r60_px	network v2	777.087	689.713	476.668	px
sholl_crossings_r10	network v2	9.1176	8	5.6886	count
sholl_crossings_r20	network v2	9.2941	10	4.4267	count
sholl_crossings_r30	network v2	9.5294	9	4.9132	count
loop_count	loops, report only	2.6471	0	4.2711	count
loop_edge_share	loops, report only	0.1141	0	0.1412	share
ring_width_mean_r30_px	width	7.4834	7.6788	1.0406	px

Per field
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measure	value	units
analysed_area_px2	986122	px^2
total_skeleton_length_px	22290	px
canalicular_length_density_per_px	0.0226037	px^-1
junction_count	494	count
junction_density_per_px2	0.00050095	px^-2
lacuna_count	20	count

lacunae_per_px2	2.028e-05	px^-2
skeleton_component_count	321	count
mean_component_length_px	69.4393	px
median_component_length_px	15	px
field_length_density_w_per_px	0.0252633	px^-1
field_density_without_flagged_per_px	0.0222092	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	603	px
unattached_component_count	31	count
canalicular_length_density_attached_per_px	0.0226037	px^-1
field_length_density_with_unattached_per_px	0.0232152	px^-1
loop_count	62	count
loop_edge_share	0.1935	share
edge_count	1292	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6174	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	17288	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	483	34.5143	18.5791	1.8577	0.8428	0.9028	-1.3729	9.6004	721.977	True
2	889	60.8664	21.5104	2.8296	0.9355	0.7566	-1.1858	41.9595	618.117	False
3	971	48.2689	39.9508	1.2082	0.5612	0.6637	-1.1736	126.652	613.53	False
4	1293	67.1123	29.3143	2.2894	0.8996	0.7085	-0.9693	147.124	738.418	False
5	933	52.1712	30.6068	1.7046	0.8098	0.7272	-0.9318	146.222	325.69	False
6	8385	126.932	101.718	1.2479	0.5982	0.7052	-1.4611	277.754	204.132	False
7	1905	74.7758	58.1104	1.2868	0.6293	0.5641	1.396	301.221	621.288	False
8	1403	54.0186	34.896	1.548	0.7633	0.888	-0.1596	368.566	862.24	False
9	3093	92.0714	54.1062	1.7017	0.8091	0.6916	1.2856	410.259	393.304	False
10	1944	82.0911	39.6724	2.0692	0.8755	0.6506	1.2002	562.526	659.101	False
11	1195	44.9336	38.8022	1.158	0.5043	0.7607	-1.4455	562.059	755.791	False
12	2018	112.423	30.231	3.7188	0.9632	0.6731	1.4432	571.53	456.488	False
13	634	48.158	30.1352	1.5981	0.78	0.5992	-1.5103	631.199	551.279	False
14	1329	62.4213	28.6247	2.1807	0.8887	0.8819	1.2484	643.479	606.14	False
15	812	43.7603	30.951	1.4139	0.7069	0.7167	-1.4386	655.233	193.442	False
16	12055	251.462	78.5981	3.1993	0.9499	0.6372	-0.5972	767.904	745.585	False
17	2798	118.207	35.7656	3.305	0.9531	0.8127	-1.3169	814.447	954.982	False
18	2396	80.3638	41.3693	1.9426	0.8573	0.8812	-1.0185	852.972	994.821	True
19	14977	374.03	83.4021	4.4847	0.9748	0.5035	-1.4304	910.923	184.857	True
20	2941	141.011	34.1041	4.1347	0.9703	0.6575	1.2599	895.57	423.196	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	1	98	161	41.1421	1	41.1421	5.5126
2	4	203	249	171.569	11	15.5971	7.3657
3	10	358	604	725.027	44	16.4779	8.6418
4	12	386	697	821.826	35	23.4807	8.1517
5	8	302	589	854.311	54	15.8206	9.3181
6	18	887	1678	2269.63	131	17.3254	7.7688
7	14	555	1208	1423.49	89	15.9943	6.9678
8	2	209	470	315.622	15	21.0415	6.8106
9	18	813	1542	3050.08	170	17.9416	7.4887
10	11	392	529	520.99	32	16.2809	7.8859
11	3	166	268	87.0122	4	21.753	5.0802
12	13	424	731	727.813	35	20.7947	7.4846
13	11	235	335	925.352	48	19.2782	6.3267
14	4	325	421	208.238	11	18.9307	7.8808
15	10	360	625	789.713	37	21.3436	8.5944
16	12	411	688	312.078	13	24.006	5.9023
17	3	66	220	153.468	7	21.924	7.6788
18	0	0	0	0	0	0	
19	20	603	1219	2081.06	117	17.7868	8.5339
20	14	432	924	1535.08	74	20.7444	7.8714

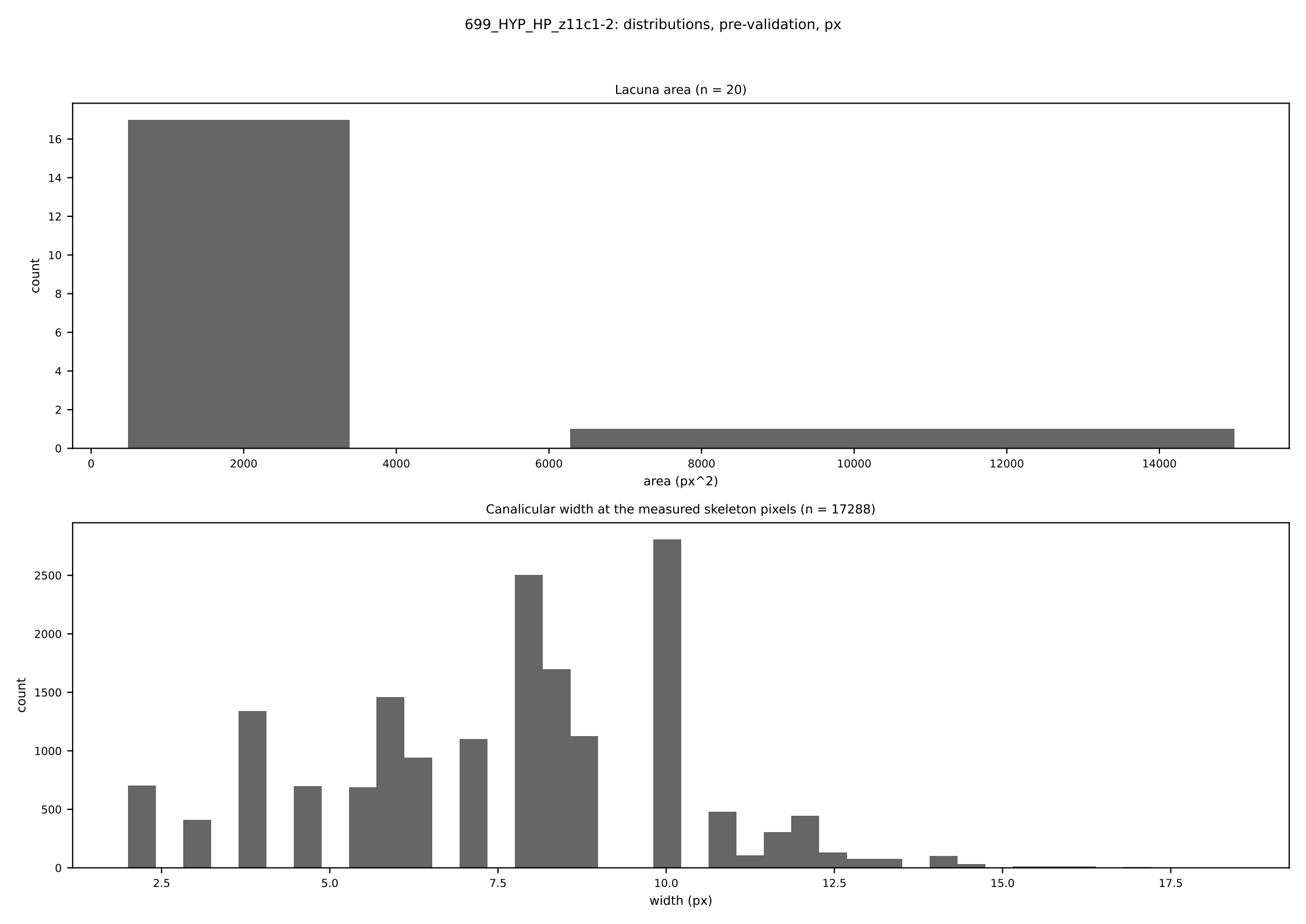
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	95.2548	3348	7313	0.6198	0.5747	0.0292712	0.0220156	1.0498
2	172.995	6815	11086	0.9811	0.8306	0.0297872	0.0224608	2.3122
3	191.622	7249	15098	1	1	0.0493861	0.0400053	5.2186
4	234.25	8068	19697	1	1	0.0478433	0.0353861	5.1227
5	172.995	7102	18424	1	1	0.0425232	0.0319692	4.6244
6	894.833	17281	36859	1	1	0.0513281	0.0455248	2.0115
7	394.333	10570	25661	1	1	0.0525071	0.0470753	3.5503
8	174.066	7351	20270	1	1	0.0284315	0.023187	1.149
9	467.446	11796	28229	1	1	0.0689217	0.0546247	3.8507
10	318.333	9236	16615	1	1	0.0424426	0.0318387	3.4555
11	232.285	6906	15617	1	1	0.0240371	0.0171608	1.2915
12	409.931	10384	22886	1	1	0.0408321	0.0319409	3.1713
13	174.853	5222	9580	1	1	0.0450019	0.0349687	6.291
14	172.267	5412	9650	1	1	0.0600517	0.0436269	2.322
15	182.894	6958	19375	1	1	0.051739	0.0322581	5.4676
16	834.833	23261	48925	1	1	0.0176691	0.0140623	1.4374
17	304.777	8057	16374	0.9164	0.8272	0.00819163	0.0134359	0.9843
18	235.48	4021	7954	0.6927	0.6449	0	0	0
19	1353.43	28693	52942	0.9016	0.8594	0.0210156	0.0230252	1.4777
20	402.333	12473	26315	1	1	0.0346348	0.035113	3.4797

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	85	99	105.184	173.64	4	2	2
2	99	103	225.924	274.38	5	6	8
3	329	496	413.99	689.713	8	8	10
4	374	621	434.049	768.115	9	12	12
5	272	545	349.907	668.215	6	9	7
6	863	1411	993.595	1886.52	25	16	17
7	526	994	621.002	1339	14	12	17
8	169	284	233.752	529.76	4	4	7

9	799	1489	916.37	1721.76	18	18	17
10	333	413	439.948	597.002	10	10	9
11	92	92	186.024	295.38	5	4	4
12	400	491	468.605	820.754	11	13	7
13	207	224	264.309	374.978	5	5	4
14	317	364	377.019	482.446	8	8	4
15	321	490	406.563	701.6	8	11	9
16	246	334	462.688	782.637	8	11	10
17	61	122	72.6985	249.066	1	1	3
18	0	0	0	0	0	0	0
19	438	1131	656.103	1351.76	15	20	15
20	271	473	479.605	1029.15	10	10	17


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 20)
count
area (px^2)
0
2
4
6
8
10
12
14
16
0
2000
4000
6000
8000
10000
12000
14000
Canalicular width at the measured skeleton pixels (n = 17288)
count
width (px)
0
500
1000
1500
2000
2500
2.5
5.0
7.5
10.0
12.5
15.0
17.5